SQL Server Always On Availability Groups

End-to-End Guide for Learning and Interviews

Who this is for: DBAs and engineers who want to understand Always On Availability Groups (AGs) from first principles, build one end to end, operate and troubleshoot it, and answer interview questions with confidence.

How to use it: Parts 1 to 3 build the theory. Part 4 is a complete lab you can follow on three Windows VMs. Parts 5 to 8 cover operations, monitoring, troubleshooting and variants. Part 9 is an interview bank with 45 questions and model answers.

Version note: Syntax shown works on SQL Server 2016 and later unless stated. Features introduced in SQL Server 2022 and 2025 are marked. Confirm limits and behavior in Microsoft Learn for your build.

Contents

- Part 1 HA and DR fundamentals (RPO, RTO, options compared)
- Part 2 AG concepts and terminology
- Part 3 Architecture and internals (commit flow, quorum)
- Part 4 End-to-end build: WSFC, endpoints, AG, listener
- Part 5 Client connectivity and read-only routing
- Part 6 Failover: planned, automatic and forced
- Part 7 Monitoring, maintenance and troubleshooting
- Part 8 Variants: Basic, Distributed, Contained, Read-scale, Linux, Azure, SQL Server 2025
- Part 9 Interview question bank (45 questions)
- Part 10 Cheat sheet and best practices

Part 1. HA and DR fundamentals

High Availability (HA) keeps a service running when a component such as a server, disk or instance fails, usually within one data center. **Disaster Recovery (DR)** restores service after a large failure such as a data center or region loss, usually in another location.

Term	Meaning	Example
RPO (Recovery Point Objective)	The maximum amount of data loss you can accept, measured in time.	RPO = 0 means no committed transaction may be lost.
RTO (Recovery Time Objective)	The maximum time the service can be down.	RTO = 2 minutes means users must be back within 2 minutes.
SLA	Contracted availability percentage.	99.99% allows about 52 minutes of downtime a year.
Failover	Moving the workload to another node or replica.	Primary replica role moves from Node1 to Node2.
Split brain	Two nodes both believe they are primary and accept writes.	Prevented by WSFC quorum and fencing.

SQL Server HA and DR options compared

Option	Unit protected	Data copies	Auto failover	Readable copy	Notes
Failover Cluster Instance (FCI)	Entire instance	1 (shared storage)	Yes	No	Storage is a single point of failure unless replicated. Protects logins, jobs, all databases.
Always On AG	Group of databases	Up to 9 replicas, each with own copy	Yes (sync replicas)	Yes	Database level. Needs instance-level objects synced separately.
Log shipping	Single database	Warm standby	No	Only in standby mode	Simple and cheap. RPO depends on backup interval.
Database mirroring	Single database	1 mirror	Yes with witness	Via snapshot	Deprecated. Replaced by AGs.
Transactional replication	Selected objects	Subscriber	No	Yes	Data distribution feature, not a true HA solution.

Interview angle: A frequent question is 'AG or FCI?'. Short answer: FCI protects the whole instance with one copy of the data on shared storage. AG protects selected databases with multiple independent copies and offers readable secondaries. Many enterprises combine them (FCI as AG replicas).

Part 2. AG concepts and terminology

An **Availability Group** is a set of user databases that fail over together. One server instance hosts the **primary replica**, which accepts read/write traffic. Up to eight **secondary replicas** receive the transaction log stream and keep their own copies of the databases in sync.

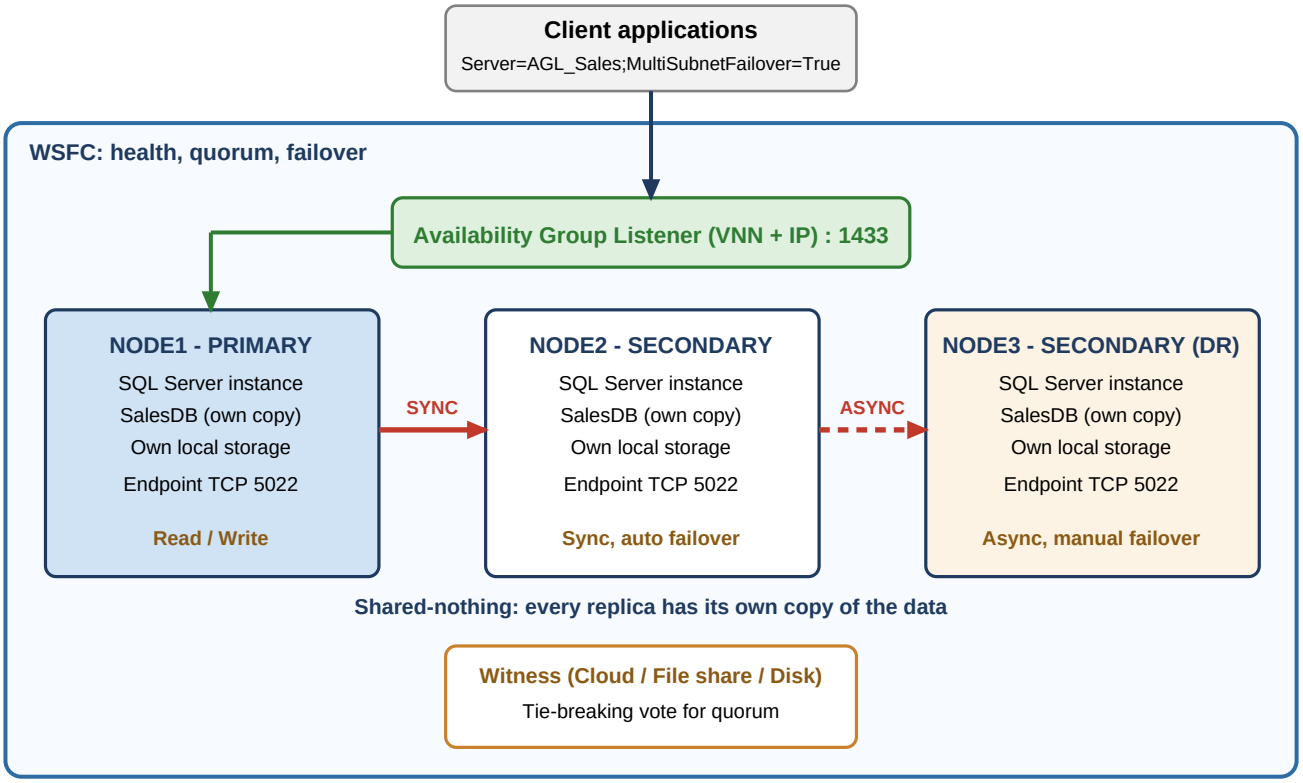
Term	Definition
Availability replica	A SQL Server instance participating in the AG. Each replica holds a copy of every database in the group.

Term	Definition
Primary replica	Hosts the read/write copy and sends log records to secondaries.
Secondary replica	Receives and hardens log records, then redoes them to its data files. Can be readable and used for backups.
Availability database	A database that is a member of the AG. It must be in FULL recovery model.
Availability mode	SYNCHRONOUS_COMMIT (primary waits for secondary to harden log) or ASYNCHRONOUS_COMMIT (primary does not wait).
Failover mode	AUTOMATIC (requires sync mode) or MANUAL (planned failover, or forced failover with possible data loss).
Availability Group Listener	A virtual network name and IP that always points to the current primary. Clients connect here, not to a node.
Database mirroring endpoint	TCP endpoint (default port 5022) used for log transport and replica communication.
WSFC	Windows Server Failover Cluster. Provides node health monitoring, quorum, and the cluster resource for the AG.
Quorum / Witness	Voting mechanism that decides if the cluster may run. A witness adds a tie-breaking vote.
Seeding	How the secondary database is initialized: AUTOMATIC (streamed over the endpoint) or MANUAL (backup and restore WITH NORECOVERY).
Log send queue	Log records on the primary not yet sent to a secondary. Indicates potential data loss (RPO).
Redo queue	Log records hardened on a secondary but not yet applied. Indicates recovery time (RTO).

Capacity limits to remember

Item	Limit
Replicas per AG (SQL Server 2014 and later)	1 primary + up to 8 secondaries (9 total)
Synchronous replicas	Up to 3 including primary in SQL Server 2016 and 2017; up to 5 in SQL Server 2019 and later
Automatic failover partners (2016 and later)	Up to 2 automatic failover partners (3 replicas in the automatic failover set)
Databases per AG	Limited by resources; planning guidance is tens of databases, depending on worker threads and workload
Edition	Enterprise for full features. Standard Edition supports only Basic Availability Groups (2 replicas, 1 database)

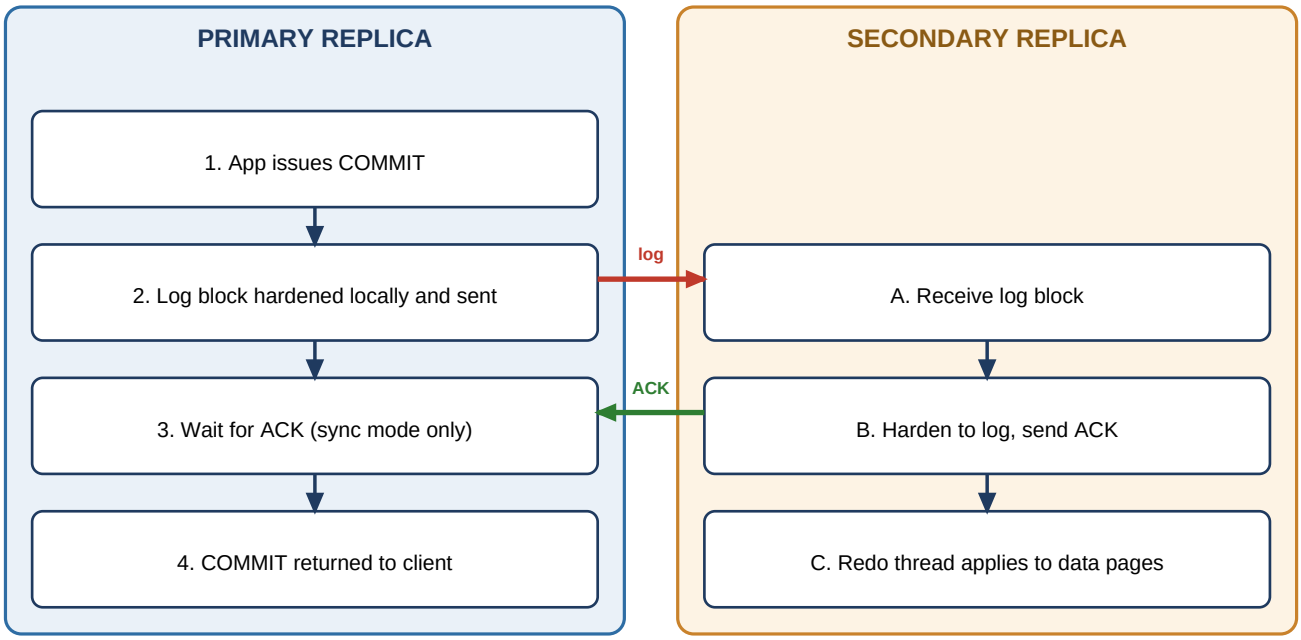
Part 3. Architecture and internals



Key architecture points

- **Shared-nothing design:** each replica uses its own storage and its own copy of the data. There is no shared disk requirement.
- **WSFC is the control plane:** it monitors health, holds quorum, and tells SQL Server which replica should be primary.
- **Log shipping at block level:** the primary sends log blocks (not transactions) over the HADR endpoint. Secondaries harden and redo them.
- **Listener is a cluster resource:** a network name plus IP address(es) that moves with the primary role.
- **Multi-subnet support:** the listener can hold one IP per subnet, enabling geographically dispersed AGs without a stretched VLAN.

How a commit flows



Mode	Commit behavior	RPO	Cost
Synchronous commit	Primary waits until the secondary has hardened the log record, then commits to the client.	Zero data loss while SYNCHRONIZED	Adds network round-trip latency to every commit (wait type HADR_SYNC_COMMIT).
Asynchronous commit	Primary commits as soon as its own log is hardened. Does not wait for secondary.	Possible data loss equal to the log send queue	Minimal impact on primary. Ideal for distant DR.

Synchronization states

State	Meaning
NOT SYNCHRONIZING	Data movement is stopped or the replica is disconnected. Check endpoint, network and suspension.
SYNCHRONIZING	Data is flowing but the secondary is catching up (always the case for async replicas).
SYNCHRONIZED	Sync secondary has hardened all log. Required for zero-data-loss and automatic failover.
REVERTING / INITIALIZING	Transient states during undo after failover or while seeding.

Quorum and witness

WSFC uses a **vote-based quorum**. The cluster stays online only while a majority of votes are reachable. This protects against split brain.

Quorum model	Votes	Use when
Node Majority	One vote per node	Odd number of nodes (3, 5).
Node and File Share Majority	Nodes + a file share witness	Even number of nodes, on-prem, third site available.
Node and Cloud Witness	Nodes + Azure Blob witness	Recommended for most modern designs, especially two-node and multi-site.

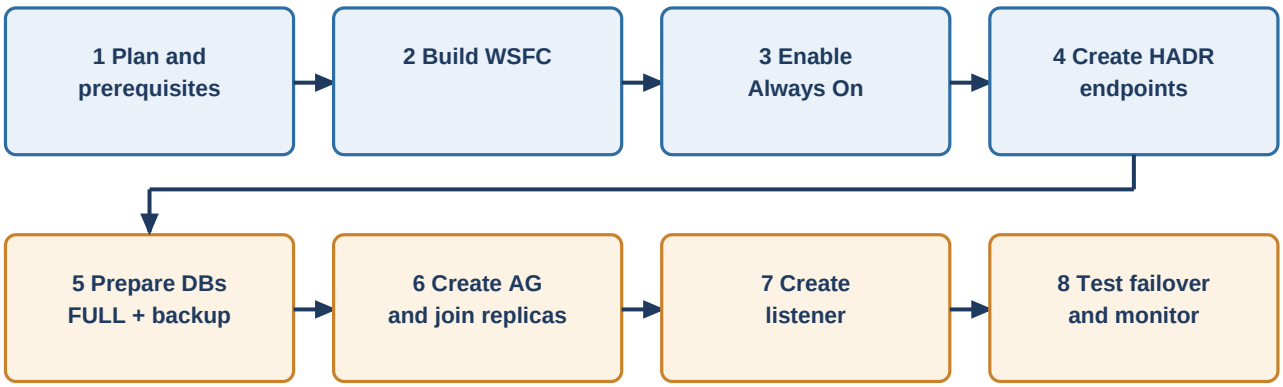
Quorum model	Votes	Use when
Node and Disk Majority	Nodes + shared disk witness	FCI environments with shared storage.

- **Dynamic quorum** adjusts vote counts as nodes leave the cluster, letting the cluster survive sequential failures.
- **Dynamic witness** turns the witness vote on or off to keep the total vote count odd.
- Give the DR node **NodeWeight = 0** when you do not want a DR outage or network split to affect primary-site quorum.
- Use **forced quorum** (start cluster with /FQ) only as a last resort during a disaster.

Key rule: Automatic failover needs **three things** together: synchronous commit on primary and target, the target database in SYNCHRONIZED state, and the WSFC in quorum. If any is missing, failover is manual.

Part 4. End-to-end build (lab)

End-to-end implementation roadmap



Row 1 builds the platform, row 2 builds and validates the Availability Group

Lab topology

Item	Value
Domain	corp.local (domain controller required)
Replicas	SQLNODE1 (10.10.1.11), SQLNODE2 (10.10.1.12), optional SQLNODE3 DR (10.20.1.13)
Cluster name / IP	SQLWSFC / 10.10.1.20
AG name / Listener	AG_Sales / AGL_Sales (10.10.1.50, port 1433)
Database	SalesDB (FULL recovery)
Service account	CORP\sqlsvc (domain account for the SQL Server service)
Witness	Cloud witness (Azure storage) or file share \\fs01\witness
Ports	1433 (SQL), 5022 (HADR endpoint), plus WSFC ports (RPC, 3343)

Prerequisite checklist: same Windows Server version and patch level on all nodes; SQL Server installed on each node with the **same edition and compatible version** (secondary must be same or higher build for upgrades); nodes in the same domain; static IPs; DNS records; firewall ports open; SQL service account with the right permissions; same drive layout or file paths for data and log files if you use manual seeding; time sync (NTP).

Step 1. Install the failover clustering feature (all nodes)

```
# Run in elevated PowerShell on SQLNODE1, SQLNODE2 (and SQLNODE3)
Install-WindowsFeature -Name Failover-Clustering -IncludeManagementTools
Restart-Computer
```

Step 2. Validate and create the cluster

```
# Validate (review the HTML report; fix errors before continuing)
Test-Cluster -Node SQLNODE1,SQLNODE2

# Create the cluster without storage (AG uses local disks)
New-Cluster -Name SQLWSFC -Node SQLNODE1,SQLNODE2 `
            -StaticAddress 10.10.1.20 -NoStorage

# Configure a cloud witness (or use -FileShareWitness \\fs01\witness)
Set-ClusterQuorum -CloudWitness -AccountName "<storageaccount>" `
                  -AccessKey "<key>"

Get-ClusterQuorum
Get-ClusterNode | Select Name, State, NodeWeight
```

Step 3. Enable the Always On feature (each instance, restarts the service)

Alternative: SQL Server Configuration Manager, SQL Server service properties, Always On High Availability tab.

```
Enable-SqlAlwaysOn -ServerInstance SQLNODE1 -Force
Enable-SqlAlwaysOn -ServerInstance SQLNODE2 -Force

-- Verify in T-SQL
SELECT SERVERPROPERTY('IsHadrEnabled') AS IsHadrEnabled;   -- 1 = enabled
```

Step 4. Create the HADR endpoint and grant permissions (each replica)

Note: SSMS wizard creates the endpoint automatically. If service accounts differ across replicas, grant CONNECT to each account.

```
CREATE ENDPOINT [Hadr_endpoint]
    STATE = STARTED
    AS TCP (LISTENER_PORT = 5022)
    FOR DATA_MIRRORING (
        ROLE = ALL,
        AUTHENTICATION = WINDOWS NEGOTIATE,
        ENCRYPTION = REQUIRED ALGORITHM AES);

GRANT CONNECT ON ENDPOINT::[Hadr_endpoint] TO [CORP\sqlsvc];

-- Verify
SELECT name, state_desc, port FROM sys.tcp_endpoints
WHERE type_desc = 'DATABASE_MIRRORING';
```

Step 5. Prepare the database on the primary

```
-- Must be FULL recovery
ALTER DATABASE SalesDB SET RECOVERY FULL;

-- A full backup is required to start the log chain
BACKUP DATABASE SalesDB TO DISK = N'\\fs01\backup\SalesDB_full.bak' WITH COMPRESSION, INIT;
BACKUP LOG      SalesDB TO DISK = N'\\fs01\backup\SalesDB_log.trn';
```

Step 6a. Create the Availability Group (run on the primary, SQLNODE1)

```
CREATE AVAILABILITY GROUP [AG_Sales]
WITH (AUTOMATED_BACKUP_PREFERENCE = SECONDARY,
      DB_FAILOVER = ON,              -- fail over if a database goes offline (2016+)
      DTC_SUPPORT = NONE)
FOR DATABASE [SalesDB]
REPLICA ON
  N'SQLNODE1' WITH (
      ENDPOINT_URL = N'TCP://SQLNODE1.corp.local:5022',
      AVAILABILITY_MODE = SYNCHRONOUS_COMMIT,
      FAILOVER_MODE = AUTOMATIC,
      SEEDING_MODE = AUTOMATIC,
      BACKUP_PRIORITY = 50,
      SECONDARY_ROLE (ALLOW_CONNECTIONS = READ_ONLY,
          READ_ONLY_ROUTING_URL = N'TCP://SQLNODE1.corp.local:1433')),
  N'SQLNODE2' WITH (
      ENDPOINT_URL = N'TCP://SQLNODE2.corp.local:5022',
      AVAILABILITY_MODE = SYNCHRONOUS_COMMIT,
      FAILOVER_MODE = AUTOMATIC,
      SEEDING_MODE = AUTOMATIC,
      BACKUP_PRIORITY = 50,
      SECONDARY_ROLE (ALLOW_CONNECTIONS = READ_ONLY,
          READ_ONLY_ROUTING_URL = N'TCP://SQLNODE2.corp.local:1433'));
```

Step 6b. Join the secondary (run on SQLNODE2)

```
ALTER AVAILABILITY GROUP [AG_Sales] JOIN;

-- Allow the AG to create the database through automatic seeding
ALTER AVAILABILITY GROUP [AG_Sales] GRANT CREATE ANY DATABASE;
```

Step 6c (alternative). Manual seeding with backup and restore

Use manual seeding for very large databases, slow or unstable networks, or when you need control over backup and restore timing.

```
-- On the secondary, restore WITH NORECOVERY, then join the database
RESTORE DATABASE SalesDB FROM DISK = N'\\fs01\backup\SalesDB_full.bak' WITH NORECOVERY;
RESTORE LOG      SalesDB FROM DISK = N'\\fs01\backup\SalesDB_log.trn'  WITH NORECOVERY;

ALTER DATABASE SalesDB SET HADR AVAILABILITY GROUP = [AG_Sales];
```

Step 7. Create the listener (on the primary)

```
ALTER AVAILABILITY GROUP [AG_Sales]
ADD LISTENER N'AGL_Sales' (
    WITH IP ((N'10.10.1.50', N'255.255.255.0')),
    PORT = 1433);

-- Multi-subnet example: add one IP per subnet
-- WITH IP ((N'10.10.1.50', N'255.255.255.0'), (N'10.20.1.50', N'255.255.255.0'))
```

Step 8. Validate the deployment

Also run the SSMS Always On Dashboard (right-click the AG, Show Dashboard) and test the listener from a client.

```
-- Replica and database health
SELECT ag.name AS ag, ar.replica_server_name, ars.role_desc,
       ars.operational_state_desc, ars.connected_state_desc, ars.synchronization_health_desc
FROM sys.availability_groups ag
JOIN sys.availability_replicas ar ON ag.group_id = ar.group_id
JOIN sys.dm_hadr_availability_replica_states ars ON ar.replica_id = ars.replica_id;

SELECT DB_NAME(database_id) AS db, synchronization_state_desc, synchronization_health_desc,
       log_send_queue_size, redo_queue_size
FROM sys.dm_hadr_database_replica_states
WHERE is_local = 1;

-- Automatic seeding progress
SELECT * FROM sys.dm_hadr_automatic_seeding ORDER BY start_time DESC;
```

Automatic seeding vs manual seeding: automatic seeding streams the database through the endpoint (2016+). It is simple and needs no shared backup path, but it uses the log transport channel, which can saturate slow WANs. Manual restore gives full control and can use differential and log backups to shorten the window.

Part 5. Client connectivity and read-only routing

Applications must connect to the **listener**, not a node. The listener follows the primary across failovers.

Connection strings

```
-- Read/write (goes to primary)
Server=tcp:AGL_Sales,1433;Database=SalesDB;Integrated Security=SSPI;MultiSubnetFailover=True;

-- Read-only intent (routed to a readable secondary if routing is configured)
Server=tcp:AGL_Sales,1433;Database=SalesDB;Integrated Security=SSPI;
ApplicationIntent=ReadOnly;MultiSubnetFailover=True;
```

- **MultiSubnetFailover=True** makes the client try all listener IPs in parallel, which cuts connection time after failover in multi-subnet setups.
- **ApplicationIntent=ReadOnly** plus a specified **Database** in the connection string are required for read-only routing.
- Set **RegisterAllProvidersIP** and **HostRecordTTL** appropriately if legacy clients cannot use MultiSubnetFailover.

Configure read-only routing

```
-- Step 1: each secondary needs a routing URL and readable role (already set in CREATE AG).
-- Step 2: define the routing list for each replica when it is primary.

ALTER AVAILABILITY GROUP [AG_Sales]
MODIFY REPLICA ON N'SQLNODE1' WITH (
    PRIMARY_ROLE (READ_ONLY_ROUTING_LIST = (N'SQLNODE2')));

ALTER AVAILABILITY GROUP [AG_Sales]
MODIFY REPLICA ON N'SQLNODE2' WITH (
    PRIMARY_ROLE (READ_ONLY_ROUTING_LIST = (N'SQLNODE1')));

-- Load-balanced routing across several secondaries (SQL Server 2016+)
-- READ_ONLY_ROUTING_LIST = (('SQLNODE2','SQLNODE3'), 'SQLNODE1')

-- Test: connect with ApplicationIntent=ReadOnly and run
SELECT @@SERVERNAME;
```

Readable secondary caveats

- Reads on a secondary use **snapshot isolation** with row versions stored in **tempdb**, so tempdb usage on the secondary rises.
- Row version overhead (14 bytes per row) is added on the primary for changed rows.
- Long-running reports on a secondary can **block the redo thread** in rare cases (for example DDL), increasing the redo queue.
- Statistics on a secondary are created in tempdb and are temporary. Consider creating needed stats on the primary.

Part 6. Failover: planned, automatic and forced

Type	Data loss	When used	Who triggers
Automatic failover	None (sync, SYNCHRONIZED)	Primary instance or database failure, node or service crash	WSFC and SQL Server health detection
Planned manual failover	None (sync, SYNCHRONIZED)	Patching, maintenance, migrations	DBA runs ALTER AVAILABILITY GROUP ... FAILOVER on the target
Forced failover	Possible	Disaster, primary unavailable, no synchronized secondary	DBA runs FORCE_FAILOVER_ALLOW_DATA_LOSS on the target

What triggers automatic failover

- **Lease timeout** (default 20 seconds): the primary must renew its lease with WSFC; failure means the primary goes offline for the AG.
- **Health check timeout** (default 30 seconds): the cluster calls sp_server_diagnostics; no response in time marks the instance unhealthy.
- **Flexible failover policy** (FailureConditionLevel, default 3): controls which conditions such as server down, unresponsive, critical errors trigger failover.
- **DB_FAILOVER = ON**: a database going offline or suspect triggers failover.
- WSFC **failover threshold** (default allows N-1 failures in a 6 hour period, where N is the replica count) stops endless failover loops; after that the AG role stays failed.

Planned manual failover (zero data loss)

```
-- 1. Confirm target is SYNCHRONOUS_COMMIT and SYNCHRONIZED (run on primary)
SELECT ar.replica_server_name, ar.availability_mode_desc, drs.synchronization_state_desc
FROM sys.availability_replicas ar
JOIN sys.dm_hadr_database_replica_states drs ON ar.replica_id = drs.replica_id;

-- 2. If the target is ASYNC, temporarily switch it to SYNC and wait for SYNCHRONIZED
ALTER AVAILABILITY GROUP [AG_Sales]
MODIFY REPLICA ON N'SQLNODE2' WITH (AVAILABILITY_MODE = SYNCHRONOUS_COMMIT);

-- 3. Connect to the TARGET secondary (SQLNODE2) and run:
ALTER AVAILABILITY GROUP [AG_Sales] FAILOVER;

-- 4. Optionally restore async mode on the DR replica afterwards.
```

Forced failover (disaster, possible data loss)

```
-- Run on the replica you want as the new primary
ALTER AVAILABILITY GROUP [AG_Sales] FORCE_FAILOVER_ALLOW_DATA_LOSS;

-- After the old primary returns, its databases are SUSPENDED. Resume deliberately:
ALTER DATABASE SalesDB SET HADR RESUME;

-- If WSFC has lost quorum, force the cluster up first (PowerShell):
-- Start-ClusterNode -Name SQLNODE2 -FixQuorum
```

Post-failover checklist

1. Confirm roles with **sys.dm_hadr_availability_replica_states** and the dashboard.
2. Verify applications reconnect through the listener and that read-only routing still works.
3. Check for **orphaned users**: logins are not replicated. Compare SIDs or use Copy-DbaLogin and sp_change_users_login or ALTER USER ... WITH LOGIN.
4. Check SQL Agent jobs: ensure jobs run only on the primary (test sys.fn_hadr_is_primary_replica in job steps).
5. Review suspended databases and the log send and redo queues.
6. Record actual RTO and RPO for your DR report.

Gotcha: The AG replicates **databases only**. Logins, SQL Agent jobs, linked servers, credentials, certificates, server-level triggers and SSIS packages in msdb are **not** replicated. Use scripts, dbatools, or a Contained Availability Group (SQL Server 2022+).

Part 7. Monitoring, maintenance and troubleshooting

Monitoring tools

Tool	Purpose
SSMS Always On Dashboard	Quick health view, failover readiness, latency and queue sizes.
DMVs	sys.dm_hadr_database_replica_states, sys.dm_hadr_availability_replica_states, sys.dm_hadr_cluster, sys.dm_hadr_automatic_seeding.
Extended Events: AlwaysOn_health	Default session capturing failovers, state changes, lease and health events. Check first when investigating.
Windows Cluster log	Get-ClusterLog -UseLocalTime -TimeSpan 30 to see WSFC decisions.

Tool	Purpose
Perfmon counters	SQLServer:Database Replica (Log Send Queue, Redo Queue), Availability Replica (Bytes Sent to Replica/sec).
SQL Agent alerts	Alerts for errors 1480 (role change), 35264/35265 (data movement suspended or resumed), 976/983 (database not accessible).

Latency and queue monitoring query

Interpretation: log_send_queue / log_send_rate approximates seconds of potential data loss (RPO); redo_queue / redo_rate approximates failover recovery time (RTO).

```
SELECT ar.replica_server_name, DB_NAME(drs.database_id) AS db,
       drs.synchronization_state_desc AS sync_state,
       drs.log_send_queue_size AS log_send_kb,
       drs.log_send_rate        AS log_send_kb_per_sec,
       drs.redo_queue_size      AS redo_kb,
       drs.redo_rate            AS redo_kb_per_sec,
       CASE WHEN drs.log_send_rate > 0
            THEN drs.log_send_queue_size * 1.0 / drs.log_send_rate END AS est_data_loss_sec,
       CASE WHEN drs.redo_rate > 0
            THEN drs.redo_queue_size * 1.0 / drs.redo_rate END         AS est_recovery_sec,
       drs.last_commit_time
FROM sys.dm_hadr_database_replica_states drs
JOIN sys.availability_replicas ar ON drs.replica_id = ar.replica_id
ORDER BY ar.replica_server_name, db;
```

Common waits to know

Wait type	Meaning
HADR_SYNC_COMMIT	Primary waiting for sync secondary to harden log. High values point to network or secondary disk latency.
HADR_DATABASE_FLOW_CONTROL	Primary throttled because the secondary is falling behind (flow control).
HADR_LOGCAPTURE_WAIT	Log capture waiting for new log to send. Usually benign idle wait.
PARALLEL_REDO_FLOW_CONTROL / REDO_THREAD_PENDING_WORK	Redo activity waits on a secondary, relevant when redo falls behind.

Troubleshooting guide

Symptom	Likely causes	What to check or do
Database NOT SYNCHRONIZING / disconnected	Endpoint stopped, firewall on 5022, service account permissions, DNS, certificate expiry	Check sys.dm_hadr_availability_replica_states.last_connect_error_description; Test-NetConnection to port 5022; endpoint STATE; GRANT CONNECT.
Data movement SUSPENDED	Manual suspend, disk full on secondary, redo failure, log chain issue	Fix cause, then ALTER DATABASE db SET HADR RESUME; check error log for 35264 and redo errors.
Primary log file keeps growing	Log cannot truncate because a secondary is behind or suspended (log_reuse_wait_desc = AVAILABILITY_REPLICA)	Resume or remove lagging replica; fix network or redo; add log space; investigate long-running transactions.
Large log send queue	Slow network, throttled bandwidth, large bulk loads, async WAN replica	Measure bandwidth, enable log stream compression (async default, optional for sync with trace flag 9592), schedule bulk operations.

Symptom	Likely causes	What to check or do
Large redo queue	Slow secondary disk, blocked redo by readable queries, CPU pressure	Check IO latency, blocking on redo thread, use parallel redo, reduce reporting load.
Listener unreachable	Cluster IP/name resource offline, DNS issue, multi-subnet client config, port conflict	Check cluster resources, DNS registration, use MultiSubnetFailover=True, verify listener port.
Automatic failover did not happen	Secondary not SYNCHRONIZED, async mode, quorum lost, failover threshold exceeded	Check availability and failover modes, quorum state, cluster log, AlwaysOn_health session.
Cluster down or quorum lost	Too many nodes or witness offline, network partition	Restore votes or start with forced quorum, then rebuild vote configuration.
Failover loop	Unhealthy primary, aggressive timeouts, resource exhaustion	Review lease and health timeouts, memory and CPU pressure, antivirus, VM snapshots and live migrations.
Automatic seeding stuck	Database not on seeding path, GRANT CREATE ANY DATABASE missing, disk or file path mismatch	Check sys.dm_hadr_automatic_seeding failure_state_desc and the error log.

Maintenance and patching

Task	Recommended approach
Rolling patch or CU	1) Patch secondary replicas (asynchronous DR first). 2) Fail over to a patched synchronous secondary. 3) Patch the old primary. 4) Optionally fail back.
OS patching	Same rolling model. Drain, pause node in WSFC, patch, resume. Check cluster validation after major changes.
Adding a database	Back up on primary, add to AG (ALTER AVAILABILITY GROUP ADD DATABASE). Automatic seeding or manual restore.
Removing a database	ALTER AVAILABILITY GROUP ... REMOVE DATABASE on the primary. Secondary copy stays in RESTORING state.
Backups	Use backup preference and sys.fn_hadr_backup_is_preferred_replica in jobs. Copy-only full and log backups on secondaries (differentials and regular full on secondary arrive with SQL Server 2025).
Index maintenance and CHECKDB	Run maintenance on the primary. Run DBCC CHECKDB on secondaries as well, because physical corruption is not replicated identically.
Job sync	Use consistent jobs on all replicas with a primary-replica check, or use contained AG msdb (2022+).

Performance considerations

- Use **fast, low-latency networking** (10 GbE where possible) and a dedicated network or QoS for HADR traffic.
- Match secondary hardware and storage performance to the primary so redo can keep pace after failover.
- **Parallel redo** (2016+) uses multiple threads per database on secondaries, improving catch-up speed.
- Reduce transaction log volume: batch size control, minimally logged operations are not available under FULL recovery so schedule bulk loads carefully.
- Keep **worker thread** capacity in mind: each database in an AG consumes threads on both primary and secondary.
- Monitor with baselines so you know normal log send and redo rates before an incident.

Part 8. Variants and modern features

Variant	What it is	Key points
Basic Availability Group	Standard Edition AG (2016+)	Two replicas, one database per AG, no readable secondary, no backups on secondary, no integrity checks on secondary.
Distributed Availability Group	An AG that spans two AGs on separate clusters	Asynchronous across sites, manual failover between AGs, good for migration and DR. Covered in our earlier document.
Contained Availability Group (2022+)	AG with its own master and msdb	Logins, permissions and SQL Agent jobs replicate with the AG, removing instance-level sync pain. SQL Server 2025 adds distributed AG support for contained AGs.
Read-scale AG (CLUSTER_TYPE = NONE, 2017+)	AG without a cluster manager	No automatic failover and no HA guarantees; used for readable scale-out and migration. Manual failover with data-loss caveats.
AG on Linux (Pacemaker)	CLUSTER_TYPE = EXTERNAL, Pacemaker as cluster manager (2017+)	Use at least three nodes or a configuration-only replica for automatic failover and quorum.
AG on Azure VMs	IaaS AG with Azure load balancer or DNN listener	Load balancer or distributed network name (DNN) for listener connectivity; cloud witness for quorum; availability zones for resilience.
AG + FCI	FCI nodes as replicas	Combines instance-level HA with database-level copies. Automatic failover for AG is not supported with FCI replicas.
Managed Instance link	AG-like replication from SQL Server to Azure SQL MI	Used for DR to Azure, read offload and low-downtime migration.

Features by SQL Server version

Version	Notable AG improvements
2012	Introduced Always On AGs: 4 replicas, 2 sync, readable secondaries, listener.
2014	Up to 8 secondaries, readable secondaries stay available during primary loss, Azure replicas.
2016	Basic AG, distributed AG, DTC support, automatic seeding, up to 3 sync replicas, DB-level health detection, parallel redo, load-balanced read-only routing.
2017	CLUSTER_TYPE (NONE and EXTERNAL), Linux support, read-scale AGs, REQUIRED_SYNCHRONIZED_SECONDARIES_TO_COMMIT.
2019	Up to 5 synchronous replicas, secondary-to-primary connection redirection, improved failover.
2022	Contained AGs, MI link, improved failover diagnostics and background tuning.
2025	Backups on secondary (full, differential, log), improved failover recovery and diagnostics, control over communication flow, REMOVE listener IP, distributed AG for contained AGs.

SQL Server 2025 reached general availability in November 2025. Verify each feature and its exact limits in the official 'What's new in SQL Server 2025' page for your build.

Part 9. Interview question bank (45 questions)

Answers are written the way you would say them in an interview: short definition first, then the detail that shows depth.

A. Fundamentals

Q1. What is an Always On Availability Group?

A feature that provides HA and DR for a set of user databases. A primary replica sends transaction log blocks to up to eight secondary replicas, each holding its own copy of the data. The group fails over as a unit and can offer readable secondaries and backup offload.

Q2. Difference between Always On AG and Failover Cluster Instance?

FCI protects the whole instance with one copy of data on shared storage and failover means the instance restarts on another node. AG protects selected databases with multiple independent copies, no shared storage, readable secondaries, and failover is a role switch. FCI protects instance-level objects; AG does not.

Q3. Is Always On the same as AG?

No. 'Always On' is a marketing umbrella covering both Availability Groups and Failover Cluster Instances. People often say Always On when they mean AG.

Q4. What are the prerequisites for an AG?

WSFC (or clusterless for read-scale), same domain for classic setup, Enterprise Edition for full features (Basic AG on Standard), FULL recovery model, a full backup, HADR enabled on every instance, endpoints, firewall ports, matching compatible SQL builds, and consistent file paths if you seed manually.

Q5. How many replicas can an AG have?

Up to 9 total (1 primary and 8 secondaries) from SQL Server 2014 onward. Up to 3 synchronous replicas in 2016 and 2017, and up to 5 in SQL Server 2019 and later. Basic AG allows two replicas.

Q6. What is a listener?

A virtual network name and IP address registered in DNS that always resolves to the current primary. Applications connect to it so they do not need to know which node is primary.

Q7. What recovery model is required?

FULL. SIMPLE recovery cannot be used because the AG depends on the transaction log.

Q8. What is the difference between synchronous and asynchronous commit?

Synchronous: primary waits for the secondary to harden the log before committing, giving RPO of zero but adding latency. Asynchronous: primary commits without waiting, lowest latency but possible data loss during failover.

Q9. What are the failover types?

Automatic (sync, synchronized, quorum), planned manual (sync, synchronized, no data loss), and forced (possible data loss, used in disasters).

Q10. Which editions support AGs?

Enterprise supports full AGs. Standard supports Basic AGs from 2016 (2 replicas, 1 database, no read access). Developer edition has Enterprise features for non-production.

Q11. What is a Basic Availability Group?

A Standard Edition feature replacing database mirroring. One database per group, two replicas, no readable secondary, no backups on the secondary.

Q12. Are logins and jobs replicated?

No. Only the databases are replicated. Sync logins with the same SIDs, jobs, linked servers and credentials using scripts, dbatools, or use a Contained AG in SQL Server 2022 and later.

B. Architecture and internals

Q13. Explain how a transaction flows to a synchronous secondary.

The client commits. The primary hardens the log locally and sends the log block over the HADR endpoint. The secondary receives and hardens it to its log, then sends an acknowledgment. The primary then returns commit to the client. The secondary's redo thread applies changes to data pages afterward, asynchronously to the commit.

Q14. What is the role of WSFC?

It monitors node and resource health, maintains quorum, hosts the AG cluster resource and listener, and coordinates failover. SQL Server uses the cluster to decide which replica becomes primary.

Q15. What is quorum and why does it matter?

Quorum is a majority vote system that determines whether the cluster can operate. It prevents split brain by ensuring only one partition runs the primary. Without quorum, the cluster stops and the AG goes offline.

Q16. What witness options exist?

Cloud witness (Azure Blob), file share witness, and disk witness. Cloud witness is recommended for modern designs, especially multi-site or two-node clusters.

Q17. What is dynamic quorum?

A WSFC feature that adjusts vote counts as nodes go offline, allowing the cluster to survive more sequential failures. Dynamic witness toggles the witness vote to keep the total odd.

Q18. Why would you set NodeWeight = 0 for a DR node?

So a DR node outage or WAN split does not reduce primary-site quorum. The DR node does not vote but can still host replicas and be forced online manually.

Q19. What is the difference between log send queue and redo queue?

Log send queue is data on the primary not yet sent to the secondary and relates to data loss (RPO). Redo queue is data hardened on the secondary but not yet applied and relates to failover time (RTO).

Q20. What are lease timeout and health check timeout?

Lease timeout (default 20 seconds) is how long the primary can go without renewing its lease before it is considered failed. Health check timeout (default 30 seconds) is how long the cluster waits for sp_server_diagnostics results. Both influence failure detection speed.

Q21. What is the HADR endpoint?

A database mirroring type TCP endpoint (default 5022) used for communication and log shipping between replicas. It supports Windows or certificate authentication and AES encryption.

Q22. What is automatic seeding and when would you avoid it?

It streams the database to secondaries over the endpoint without manual backup and restore (2016+). Avoid it for very large databases over slow or unreliable WANs, since it competes with log traffic and can stall.

Q23. How do readable secondaries work internally?

They use snapshot isolation with row versions in tempdb. Redo continues while readers query a consistent point-in-time view, but tempdb pressure and rare redo blocking can occur.

Q24. What is DB_FAILOVER?

A 2016+ AG option that triggers failover when a database in the AG goes offline, in addition to instance-level health checks. Without it, only instance-level failures trigger automatic failover.

C. Operations and troubleshooting

Q25. How do you perform a planned failover with no data loss?

Ensure the target is a synchronous replica in SYNCHRONIZED state. Connect to the target and run ALTER AVAILABILITY GROUP [AG] FAILOVER. Then verify roles, connectivity, jobs and logins.

Q26. When do you use FORCE_FAILOVER_ALLOW_DATA_LOSS?

Only when the primary is unavailable and no synchronized secondary can do a planned failover. Accept possible data loss, run it on the chosen secondary, then handle suspended databases on the old primary after it returns.

Q27. A database shows NOT SYNCHRONIZING. What do you do?

Check connectivity and endpoint (port 5022, state, permissions), the replica connected_state and last_connect_error_description, whether data movement is suspended, disk space on the secondary, and the error logs. Resume data movement if suspended.

Q28. The log file on the primary is growing and not truncating. Why?

log_reuse_wait_desc probably shows AVAILABILITY_REPLICA: a secondary is behind or disconnected so the log cannot be reused. Fix connectivity or redo lag, resume movement, or remove the replica if it will not return soon, then take log backups.

Q29. Automatic failover did not happen. What do you check?

Whether both replicas were synchronous with automatic failover mode, the secondary was SYNCHRONIZED, the cluster had quorum, the failover threshold was not exceeded, and DB_FAILOVER and failure condition level matched the event. Review AlwaysOn_health and the cluster log.

Q30. How do you handle backups in an AG?

Set the backup preference (for example prefer secondary) and use sys.fn_hadr_backup_is_preferred_replica in the backup job. Secondaries support copy-only full and log backups before SQL Server 2025, which adds full, differential and log backups on secondaries.

Q31. How do you patch an AG with minimal downtime?

Rolling approach: patch secondaries first, fail over to a patched sync secondary, patch the former primary, and optionally fail back. Keep async DR patched first and confirm health between steps.

Q32. How do you add a database to an existing AG?

Ensure FULL recovery, take a full backup, run ALTER AVAILABILITY GROUP ADD DATABASE on the primary, and let automatic seeding or manual restore WITH NORECOVERY and ALTER DATABASE SET HADR AVAILABILITY GROUP finish the join on secondaries.

Q33. What breaks after failover that DBAs often forget?

Orphaned users (SID mismatch), missing logins, SQL Agent jobs, linked servers, credentials, certificates, and jobs that should only run on primary. Plan sync of instance-level objects or use a contained AG.

Q34. How do you monitor AG health?

Dashboard, DMVs for sync state and queues, AlwaysOn_health Extended Events, cluster logs, Perfmon, and alerts on errors like 1480, 35264 and 35265. Track log send and redo rates as baselines.

Q35. What is a split brain and how does WSFC prevent it?

Two partitions both think they own the primary role. Quorum ensures only the partition with majority votes can run resources, and the minority side shuts down its AG role.

Q36. How can you reduce failover time?

Keep redo queue small (fast secondary storage, parallel redo), tune timeouts carefully, minimize long-running transactions, use accelerated database recovery to shorten undo, and keep client retry logic and MultiSubnetFailover enabled.

D. Scenario-based questions

Q37. Your DR replica in another region keeps falling behind during month-end batch loads. What do you do?

Confirm it is async. Check bandwidth and the log send queue. Options: increase bandwidth or enable compression, stagger or batch loads, size the redo side, and use log send metrics to set alerting thresholds. Accept a higher temporary RPO or reschedule heavy jobs.

Q38. Primary site is down and the cluster lost quorum. How do you recover?

Confirm the primary is truly unavailable. Start the DR node with forced quorum, then run FORCE_FAILOVER_ALLOW_DATA_LOSS on the DR replica. Repoint DNS or listener, verify applications, document data loss, and rebuild the old primary as a secondary later after reviewing differences.

Q39. Users complain about slow commits after you added a sync replica in a remote data center. What happened?

Synchronous commit now waits for the remote round trip. Check HADR_SYNC_COMMIT waits and latency. Make the remote replica asynchronous, or place sync replicas in the same low-latency site and keep the remote one async.

Q40. After failover, applications cannot connect. Where do you look?

Listener resources and DNS, client MultiSubnetFailover setting, firewall, listener port, cluster IP online state, and application connection strings that point to node names instead of the listener. Also check logins missing on the new primary.

Q41. Design an AG for a bank with RPO 0 locally and a DR site 500 km away.

Two synchronous replicas in the primary data center with automatic failover, one asynchronous replica in DR with manual failover. Cloud witness at a third location, NodeWeight 0 on DR, listener with an IP per subnet, regular DR drills, and monitoring on queues. Consider a distributed AG if the DR site has its own cluster.

Q42. How would you migrate a 2 TB database to new hardware with minimal downtime using an AG?

Build the new server and join it as a secondary using backup and restore (or seeding), let it catch up, switch to synchronous when ready, perform a planned failover during a small window, update connections via the listener or DNS, then remove the old replica.

Q43. A secondary used for reporting causes a large redo queue. What do you do?

Check for long reports blocking redo and DDL, evaluate tempdb and IO, move heavy reporting to a dedicated async replica, tune queries, and monitor redo rate. Consider read-only routing to spread load among multiple secondaries.

Q44. How do you avoid orphaned users after failover?

Create logins with the same SIDs on all replicas, schedule login sync, use a contained AG or contained database users, and run an orphan check as part of the failover runbook.

Q45. Your cluster fails over repeatedly every few hours. What is your approach?

Review the AlwaysOn_health session, cluster log and Windows event logs for the trigger. Look at CPU or memory pressure, long GC or VM migration pauses, antivirus, timeouts too aggressive, and network drops. Fix root cause before touching thresholds. Confirm the failover threshold to avoid role failure.

Q46. What would you tell a manager asking if AG replaces backups?

No. AGs protect against infrastructure failure, not logical errors such as accidental deletes or corruption that is replicated. You still need regular full, differential and log backups, tested restores, and point-in-time recovery.

Part 10. Cheat sheet and best practices

Command quick reference

Task	Command
Check HADR enabled	`SELECT SERVERPROPERTY('IsHadrEnabled');`
Who is primary?	`SELECT sys.fn_hadr_is_primary_replica('SalesDB');`
Planned failover (on target)	`ALTER AVAILABILITY GROUP [AG] FAILOVER;`
Forced failover	`ALTER AVAILABILITY GROUP [AG] FORCE_FAILOVER_ALLOW_DATA_LOSS;`
Suspend / resume movement	`ALTER DATABASE db SET HADR SUSPEND;` / `... RESUME;`
Add database	`ALTER AVAILABILITY GROUP [AG] ADD DATABASE db;`
Remove database	`ALTER AVAILABILITY GROUP [AG] REMOVE DATABASE db;`
Change commit mode	`ALTER AVAILABILITY GROUP [AG] MODIFY REPLICA ON 'N2' WITH (AVAILABILITY_MODE = ASYNCHRONOUS_COMMIT);`
Change failover mode	`... WITH (FAILOVER_MODE = MANUAL);`
Seeding progress	`SELECT * FROM sys.dm_hadr_automatic_seeding;`
Cluster quorum	`Get-ClusterQuorum; Get-ClusterNode`
Cluster log	`Get-ClusterLog -UseLocalTime -TimeSpan 30`

Best practices summary

1. Use **synchronous commit with automatic failover** inside the primary site and **asynchronous** for remote DR.
2. Use a **cloud witness** or third-site witness and plan votes (NodeWeight) deliberately.
3. Connect applications to the **listener**, use **MultiSubnetFailover=True**, and implement retry logic.
4. Match **hardware and storage** on replicas, especially if any may become primary.
5. Keep **logins, jobs and linked servers** in sync, or adopt a contained AG on 2022 and later.
6. Monitor **log send queue, redo queue, and sync state** with alerts; baseline normal rates.
7. **Back up from secondaries** using backup preference, and test restores regularly.
8. Patch with the **rolling method**: secondaries first, then fail over, then the former primary.
9. Document and rehearse **failover and DR runbooks**; record real RTO and RPO.
10. Secure the endpoint: encryption required, restricted permissions, and firewall rules limited to replicas.

Interview tips

- Lead with a one-line definition, then add architecture, then a real example. Interviewers want to see structured thinking.
- Use numbers: default timeouts (20s lease, 30s health), port 5022, replica limits, RPO and RTO.
- Always mention what is **not** replicated (logins, jobs) and why AG is not a backup replacement.
- Prepare one real incident story: symptom, diagnosis path with DMVs and logs, fix, and prevention.

- Be ready to draw the architecture: WSFC, replicas, listener, endpoint, witness, and the commit flow.